Multimedia Signal Processing - 2nd Module

September 6, 2024

1. **Question (7 pts)**
   Consider a continuous source X that emits symbols following a probability distribution function (p.d.f.) $p_X(x)$.
   (a) Consider the following series of symbols emitted by X

   $$x = \{-0.8,\ 0.1,\ -0.8,\ -0.7,\ 0.7,\ -0.7,\ -0.1,\ 0.2,\ 0.8,\ -0.2,\ 1.2\}.$$

   Design a Lloyd-Max quantizer tailored to these symbols using $M = 3$ bins and the thresholds $\{-\infty, -0.5, 0.3, +\infty\}$ as initialization. Stop after one complete iteration.
   (b) How many iterations must be performed to reach the algorithm's convergence (if possible)?
   (c) Quantize x using the Lloyd-Max quantizer designed at point (b), thus obtaining $\tilde{x}$.
   (d) Compute the entropy of the source representing $\tilde{x}$ estimating the probability mass function (p.m.f.) directly from $\tilde{x}$.

2. **Question (5 pts)** Consider the problem of video coding.
   (a) With the help of a block diagram, describe the typical hybrid DPCM/DCT video coding scheme (encoder and decoder).
   (b) Quickly explain why motion compensation is useful in this scheme.

3. **Question (3 pts)** Consider the problem of transform coding.
   (a) Explain the principle of Transform Coding.
   (b) Specify which family of transforms one should use.

4. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Consider two independent discrete sources X and Y with entropy $H(X) = 1.5$ and $H(Y) = 1.5$. The joint entropy must be $H(X, Y) = 1.5$.
   - [T] [F] A PCM scheme may be preferable to a DPCM scheme in terms of rate-distortion in some cases.
   - [T] [F] The absolute threshold of hearing can be used for bit allocation in audio coding.
   - [T] [F] In a video coding scheme, motion vectors can represent motion only in terms of integers due to pixel discretization of a frame.
   - [T] [F] Considering a transform coding scheme, it always provide better rate-distortion results of a PCM scheme.
   - [T] [F] A scalar uniform quantizer and a Lloyd-Max quantizer provide the same solution for a uniformly distributed source.
   - [T] [F] When using linear predictive vocoder exploiting the source-system model, the coefficients of the synthesis filter are directly quantized.
   - [T] [F] The absolute threshold of hearing shows that the minimum sound pressure level (SPL) needed to hear a sound at 100 Hz is lower than the SPL needed to hear a sound at 4000 Hz.

5. **Question (Matlab - 10pts)**

   Consider you have an image file called `peppers.png` in your Matlab working directory. This is a RGB image with resolution 512×384.

   Data loading

   (a) Load the image and display it.

   (b) Crop the top-left 256×256 pixels of the image (use the cropped version of the image hereinafter).

   (c) Convert to grayscale and display the image again. Feel free to chose the RGB-to-gray conversion you want.

   DCT-based coding

   Considering the color component you extracted from the image, implement a 2D transform coding technique based on 2D DCT. Use a block size of 8x8, and a quantizer that works on 16 levels.

   (d) Split the image into blocks.

   (e) Encode and decode each block using 2D DCT and the correct quantizer.

   (f) Explain how you chose the quantization step.

   (g) Reconstruct the decoded image and display it.

   (h) Compute the PSNR.